

Genome analysis

DEploid-BEST: an optimised workflow for deconvolving mixed *P. falciparum* infections

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Abstract

Summary: As a state-of-the-art method for malaria parasite genome deconvolution, the program DEploid takes read counts from a mixed sample and estimates the number of strains, their relative proportions and their sequences from a set of reference strains. This output provides a unique opportunity for researchers to study the epidemiology of infectious diseases. However, the complexity of DEploid is $\mathcal{O}(lm^2)$, where m is the number of reference strains and l is the number of sites, so it does not scale well to reference panels with many strains. To address this, we present DEploid-LASSO, which can take thousands of reference strains and shrink and optimise the panel, together with a revised pipeline that we call DEploid-BEST, the latest best practice for deconvolving unknown mixed genomes.

Availability and Implementation: The open source implementation is freely available at <https://github.com/DEploid-dev/DEploid> under the conditions of the GPLv3 license. DEploid-BEST is implemented in C++ with an R interface.

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Supplementary information: Supplementary data are available at *Bioinformatics* online.

1 Introduction

Deconvolution of mixed parasite genomes is critical for understanding malaria epidemiology, but it is challenging: the number of strains and their abundances are unknown, sequencing is error-prone, reference strains may be lacking, and recombination rates vary widely. Moreover, mixed parasite genomes arise in different ways: super-infections are transmitted by multiple mosquitoes, whereas co-infections are transmitted by a single mosquito, in which parasites undergo sexual recombination. Distinguishing these scenarios is difficult but crucial for mapping malaria transmission, since super-infections reflect transmission intensity while co-infections do not.

A mixed infection can be completely described by the number of co-existing strains K , their relative proportions $\mathbf{w}$, within-sample relatedness r and their associated haplotypes $\mathbf{h}$. These four variables describe different biological scenarios and can be studied jointly (Galinsky *et al.*, 2015; O'Brien *et al.*, 2016; Henden *et al.*, 2018) or independently (Chang *et al.*,

2017). To our knowledge, DEploid (Zhu *et al.*, 2018, 2019) is the only software package that can examine all four parameters under the same framework. DEploid uses imbalanced allele frequencies to infer the number of strains and their relative proportions, conditions on haplotype structures within a reference panel to infer strain sequences, and estimates genome-wide in-sample relatedness via identity-by-descent (IBD) states S between mixed strains. These outputs allow researchers to study mixed malaria infections with an unprecedented level of detail (Auburn *et al.*, 2019).

2 Implementation

Zhu *et al.* (2018, 2019) conducted a series of simulation studies and validations on both *in vitro* and *in silico* mixed genomes to mimic real-world epidemiological scenarios, and the software was then applied to one of the largest malaria parasite genome research projects (Pf3k Consortium, 2016) to deconvolve mixed parasite genomes. Here we summarise the technical lessons learned in the process:

1. Inferring the number of strains K is trivial in comparison to the other parameters, so K should be fixed first.
2. Once K is fixed, inferring strain proportions $\mathbf{w}$ becomes easier. Because strains can be related within a sample, this forms structured allele-frequency imbalance; it is therefore helpful to infer $\mathbf{w}$ and the within-sample relatedness structure $\mathcal{S}$ at the same time.
3. Given fixed K and $\mathbf{w}$, it becomes straightforward to infer haplotypes $\mathbf{h}$ with an appropriate reference panel.

These execution orders follow from the posterior decomposition of the model, which we recall in Section S2.1 of the Supplementary Material.

A reference panel should ideally be as diverse as possible to capture the key genomic backbones of the mixed parasites, and an appropriate panel is essential for ensuring high haplotype quality. In the case of balanced mixtures, a good reference panel can tease mixed strains apart, whereas a less appropriate panel may lead to underestimating K . Inferring the number of strains and relative abundance does not require whole-genome data: several hundred high-quality single-nucleotide polymorphism (SNP) sites are sufficient, and in the extreme case it is possible to gain quick insights by using only the longest chromosome (14) to infer K .

In light of this experience, we propose (I) a novel approach with a LASSO algorithm to optimise the reference panel and (II) an optimised workflow for mixed-genome deconvolution.

2.1 Optimised reference panel with LASSO

Let y denote the within-sample allele frequency of the mixed sample, and let X be the genotypes of the reference strains. Assuming the genotypes are independent covariates, we fit the linear model $y = \beta_0 + \vec{\beta} X$, where $\vec{\beta}$ are the weights of the reference strains that contribute towards the estimated within-sample allele frequency of the mixed genome. A key feature of the LASSO approach is that $\vec{\beta}$ is sparse, producing zero weights for many reference strains in the panel and thereby reducing its size (Friedman *et al.*, 2010). In practice we search over the penalty λ between 0 and 1, adding reference strains into the model one at a time while each addition reduces the likelihood deviance by at least one percent, and we further reduce the panel by discarding weights below 0.01. The full model and algorithm are given in the Supplementary Material.

2.2 Optimised work-flow

In the first attempt at deconvolving mixed genomes, Zhu *et al.* (2018) solved K , $\mathbf{w}$ and $\mathbf{h}$ at the same time; this method struggles to recover the correct strain proportions when mixed strains are closely related. The problem was addressed by Zhu *et al.* (2019) with an enhanced hidden Markov model for within-sample relatedness, together with an improved pipeline that separates the haplotype-inference step from the inference of K , $\mathbf{w}$ and within-sample relatedness. Nonetheless, the lack of an appropriate reference-panel selection procedure constrains haplotype inference and its usage.

The core of DEploid is Markov chain Monte Carlo (MCMC). In previous versions we recommended repeating the deconvolution 15 times with different random seeds and taking the consensus answer at the end. In the new version, DEploid-BEST executes the deconvolution algorithms in an optimised sequence to best report the number of strains, proportions and haplotypes. The program DEploid-LASSO learns the number of strains with an optimised reference panel. DEploid-IBD then fixes the number of strains and tunes the strain proportions with an identity-by-descent model. Finally, DEploid-LASSO fixes the number of strains and proportions, and uses the optimised reference panel again to infer and report the haplotypes. The optimised execution order is summarised in Figure 1.

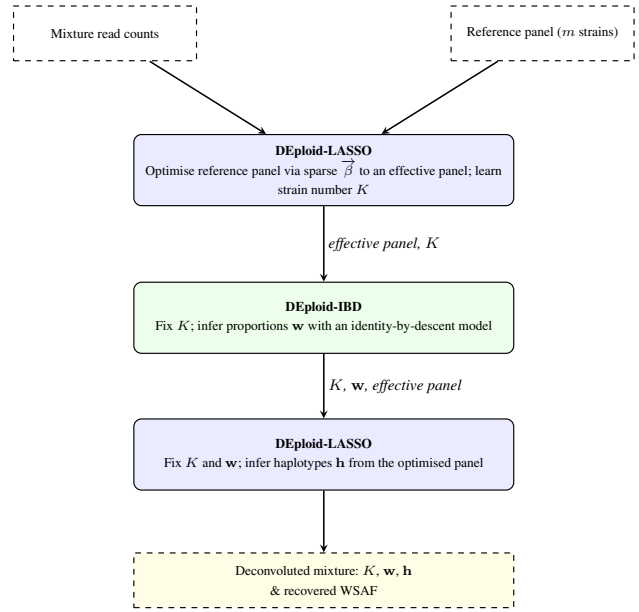


Fig. 1. The DEploid-BEST workflow. Given mixture read counts and a reference panel of m strains, DEploid-LASSO first optimises the panel (sparse weights $\vec{\beta}$, keeping $\beta_j \geq 0.01$) and learns the strain number K automatically; DEploid-IBD then fixes K and infers the proportions $\mathbf{w}$ under an identity-by-descent model; DEploid-LASSO finally fixes K and $\mathbf{w}$ and infers the haplotypes $\mathbf{h}$ from the optimised panel. The output is the deconvoluted mixture ($K, \mathbf{w}, \mathbf{h}$) and the recovered within-sample allele frequency (WSAF). The execution order follows the posterior decomposition (Section S2.1).

3 Results

On four *in-silico* mixtures of four known lab strains with known proportions (including a balanced 0.25/0.25/0.25/0.25 mixture), DEploid-BEST recovered the correct number of strains ($K = 4$) automatically in every case, whereas the classic pipeline required K to be supplied and DEploid-IBD collapsed to $K = 2$ even when $K = 4$ was given. For proportion recovery, DEploid-BEST matched a classic run given the true K (mean total absolute error 0.35 for both; IBD 1.06), the residual error reflecting an inherent 3D7/7G8 labelling ambiguity among the closely related reference strains. Full per-strain results are in Supplementary Table S2.

On realistic *in-silico* mixtures built from genuine field haplotypes (two field populations, five mosaic scenarios, four known proportion sets), DEploid-BEST recovers the mixture allele frequency with per-site Pearson r of 0.97–0.99 without any strain number being supplied, matching or slightly exceeding the fixed- $K = 4$ classic and IBD pipelines (Supplementary Table S3).

DEploid-BEST also reduces run time. On a single thread (Intel Core i9-14900HX, 31 GB RAM) deconvolving $l = 18,570$ sites, the classic and IBD pipelines scale approximately quadratically with reference-panel size m and fail to complete within a 5-minute budget once m exceeds roughly 20–50 strains, whereas DEploid-BEST completes in roughly 2–3 minutes regardless of panel size, including $m = 1000$ (Supplementary Table S1).

To make the package more accessible and to reduce the complexity of installation under different environments, we have moved the stable, generalised code into C++ libraries and built an R interface that allows users to interact directly with DEploid results without saving them to text files and reloading them for analysis. See the Supplementary Material for details.

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Conflict of Interest

Software development and data analysis in this work was completed by S.J.Z. during a postdoc at the BDI. S.J.Z. is a principal data scientist at Roche. There is no conflict of interest.

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